Agile Lab – Fat

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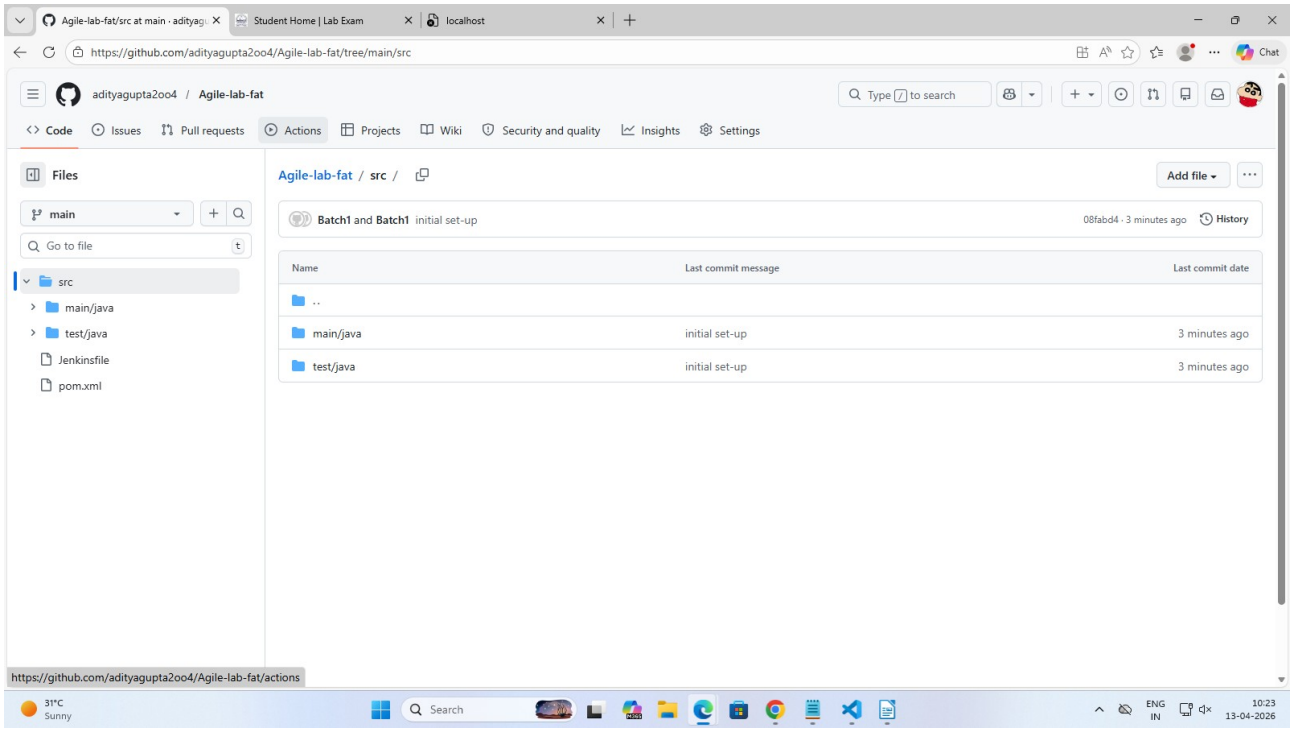
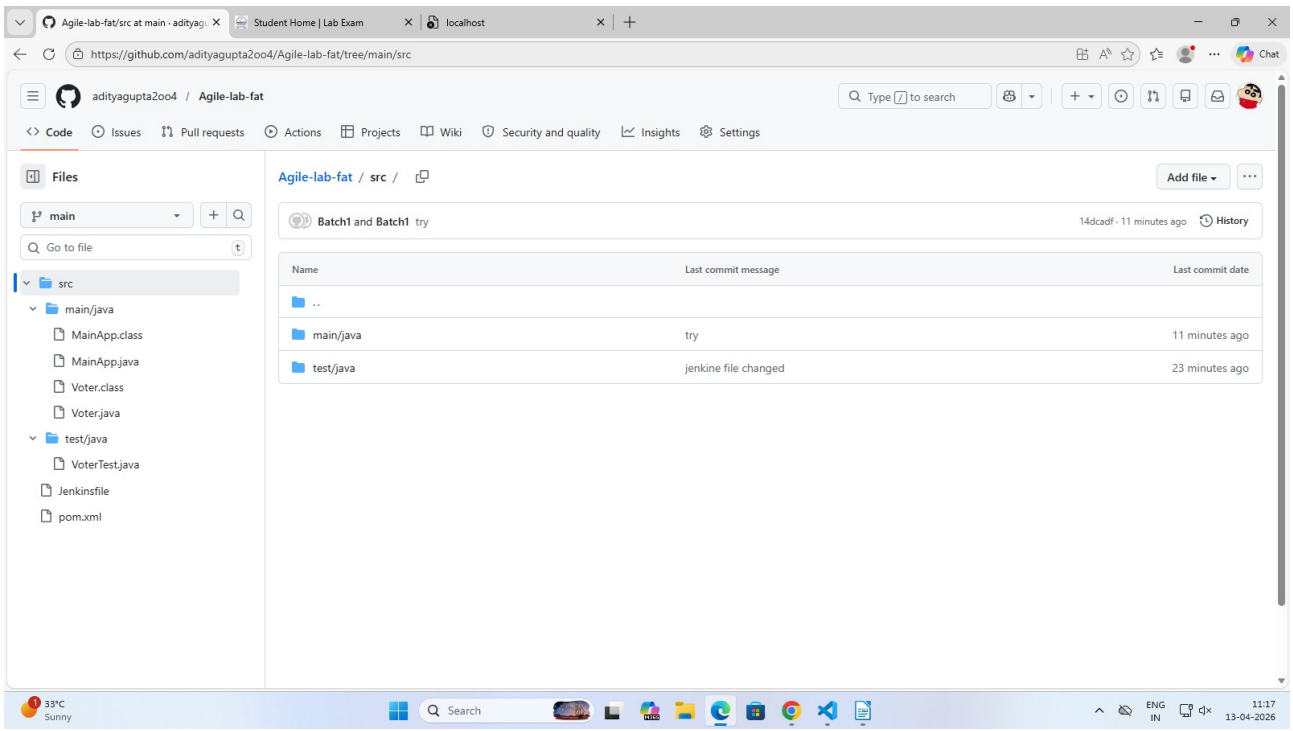
Design and implement a Voting Eligibility System program that determines whether a person is eligible to vote based on specific criteria. The program should define an appropriate class to store voter details such as name, age, citizenship, voter ID, and ID validity status. It must accept input for one or more voters and evaluate eligibility using the conditions that the voter must be at least 18 years old, must be a citizen of the country (e.g., Indian citizen), and must possess a valid voter ID. The program should display whether each voter is eligible or not, and in cases of ineligibility, it should clearly indicate the reason, such as being underage, not a citizen, or having an invalid ID

• Create a GitHub repository for a Maven-based application following the standard Maven directory structure [10]





• Clone this repository into VS Code, make the necessary code modifications, and commit the changes back to the Git repository [10]

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL

PS C:\Users\batch1\Downloads\Aditya> git clone https://github.com/adityagupta2oo4/Agile-lab-fat.git
Cloning into 'Agile-lab-fat'...
remote: Enumerating objects: 11, done.
remote: Counting objects: 100% (11/11), done.
remote: Compressing objects: 100% (6/6), done.
remote: Total 11 (delta 0), reused 11 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (11/11), done.
PS C:\Users\batch1\Downloads\Aditya>
```

```
PS C:\Users\batch1\Desktop\Aditya\Agile-lab-fat> git add .
 create mode 100644 src/test/java/VoterTest.java
PS C:\Users\batch1\Desktop\Aditya\Agile-lab-fat> git push origin main
info: please complete authentication in your browser...
Enumerating objects: 11, done.
Counting objects: 100% (11/11), done.
Delta compression using up to 20 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (11/11), 1.30 KiB | 443.00 KiB/s, done.
Total 11 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/adityagupta2oo4/Agile-lab-fat.git
 * [new branch]      main -> main
```

```
PS C:\Users\batch1\Desktop\Aditya\Agile-lab-fat> git add .
PS C:\Users\batch1\Desktop\Aditya\Agile-lab-fat> git commit -m "final commit"
[main 3d345af] final commit
 Committer: Batch1 <Batch1@VITUNIVERSITY.LOCAL>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly. Run the
following command and follow the instructions in your editor to edit
your configuration file:

    git config --global --edit

After doing this, you may fix the identity used for this commit with:

    git commit --amend --reset-author

 1 file changed, 18 insertions(+)
```

```
PS C:\Users\batch1\Desktop\Aditya\Agile-lab-fat> git push origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 20 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 437 bytes | 437.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/adityagupta2oo4/Agile-lab-fat.git
   08fabd4..3d345af  main -> main
PS C:\Users\batch1\Desktop\Aditya\Agile-lab-fat>
```

• Build and verify the application within VS Code to ensure it is compatible with Jenkins integration [10]

```
PS C:\Users\batch1\Desktop\Aditya\Agile-lab-fat\src\main\java> java MainApp
=== Voting Eligibility Checker ===
Enter Voter Name: Aditya
Enter Age: 22
Enter Citizenship (e.g., Indian): Indian
Does the voter have a valid ID? (true/false): true

--- Result for Aditya ---
Eligible to vote!
PS C:\Users\batch1\Desktop\Aditya\Agile-lab-fat\src\main\java>
```

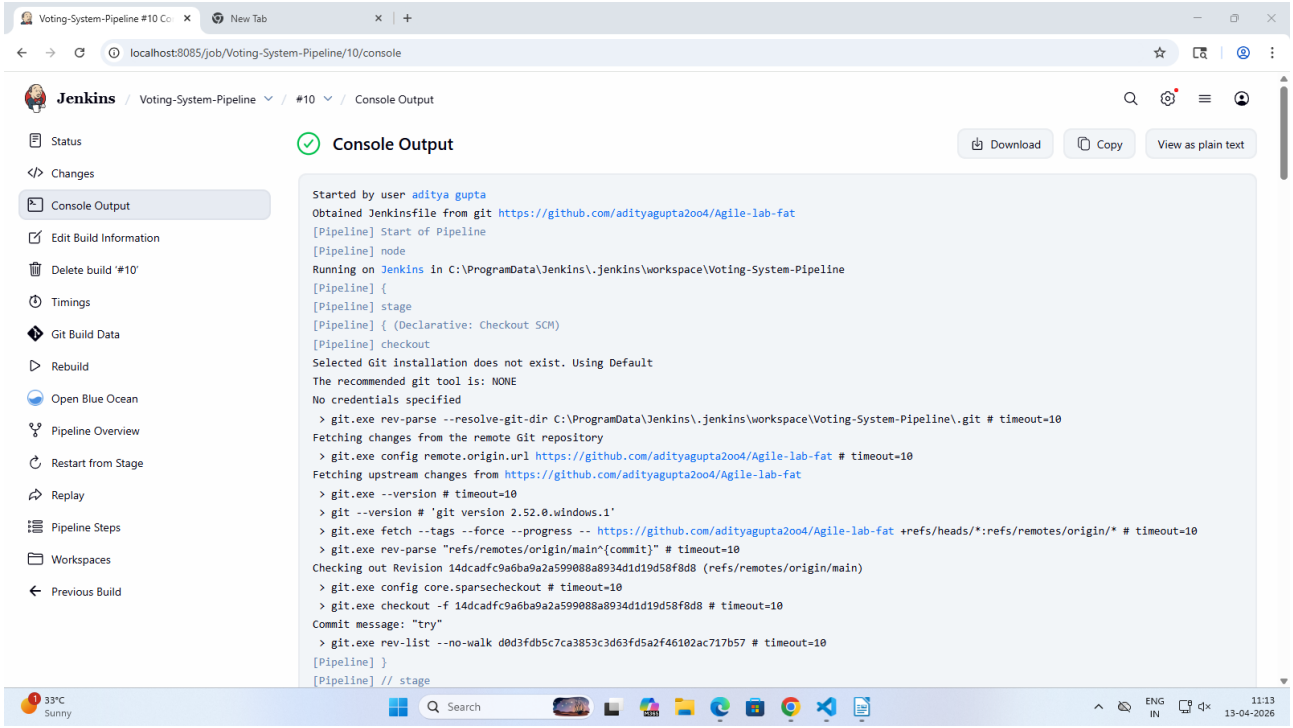
```
PS C:\Users\batch1\Desktop\Aditya\Agile-lab-fat\src\main\java> java MainApp
=== Voting Eligibility Checker ===
Enter Voter Name: Aditya
Enter Age: 17
Enter Citizenship (e.g., Indian): Indian
Does the voter have a valid ID? (true/false): false

--- Result for Aditya ---
Ineligible: Underage (Must be 18+).
PS C:\Users\batch1\Desktop\Aditya\Agile-lab-fat\src\main\java>
```

```
PS C:\Users\batch1\Desktop\Aditya\Agile-lab-fat> mvn compile
[INFO] Scanning for projects...
[INFO]
[INFO] -----------------------< com.voting:VotingSystem >-----------------------
[INFO] Building VotingSystem 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ VotingSystem ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\batch1\Desktop\Aditya\Agile-lab-fat\src\main\resources
[INFO]
[INFO] --- compiler:3.13.0:compile (default-compile) @ VotingSystem ---
[INFO] Recompiling the module because of changed source code.
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 2 source files with javac [debug target 17] to target\classes
[WARNING] system modules path not set in conjunction with -source 17
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  0.960 s
[INFO] Finished at: 2026-04-13T10:41:34+05:30
[INFO] ------------------------------------------------------------------------
PS C:\Users\batch1\Desktop\Aditya\Agile-lab-fat>
```

```
PS C:\Users\batch1\Desktop\Aditya\Agile-lab-fat> mvn test
[INFO] --- resources:3.3.1:resources (default-resources) @ VotingSystem ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\batch1\Desktop\Aditya\Agile-lab-fat\src\main\resources
[INFO]
[INFO] --- compiler:3.13.0:compile (default-compile) @ VotingSystem ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- resources:3.3.1:testResources (default-testResources) @ VotingSystem ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\batch1\Desktop\Aditya\Agile-lab-fat\src\test\resources
[INFO]
[INFO] --- compiler:3.13.0:testCompile (default-testCompile) @ VotingSystem ---
[INFO] Recompiling the module because of changed source code.
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 17] to target\test-classes
[WARNING] system modules path not set in conjunction with -source 17
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ VotingSystem ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running VoterTest
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.040 s -- in VoterTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  1.737 s
[INFO] Finished at: 2026-04-13T10:41:57+05:30
[INFO] ------------------------------------------------------------------------
```

Do you want to inst

• Finally, set up a Pipeline project in Jenkins and execute the CI/CD process by writing and running the required pipeline script [20]



Voting-System-Pipeline #10 Co | New Tab

localhost:8085/job/Voting-System-Pipeline/10/console

Jenkins / Voting-System-Pipeline / #10 / Console Output

```
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running VoterTest
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.040 s -- in VoterTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jar:3.5.0:jar (default-jar) @ VotingSystem ---
[INFO] Building jar: C:\ProgramData\Jenkins\.jenkins\workspace\Voting-System-Pipeline\target\VotingSystem-1.0-SNAPSHOT.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  1.652 s
[INFO] Finished at: 2026-04-13T11:06:33+05:30
[INFO] ------------------------------------------------------------------------
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
```

33°C Sunny | Search | ENG IN | 11:14 13-04-2026

Voting-System-Pipeline Config | New Tab

Voting-System-Pipeline - Jenki | New Tab

localhost:8085/job/Voting-System-Pipeline/

Jenkins / Voting-System-Pipeline

- Changes
- Build Now
- Configure
- Delete Pipeline
- Full Stage View
- Favorite
- Job Config History
- Open Blue Ocean
- Stages
- Rename
- Pipeline Syntax
- Set Next Build Number

Builds

Filter

Today

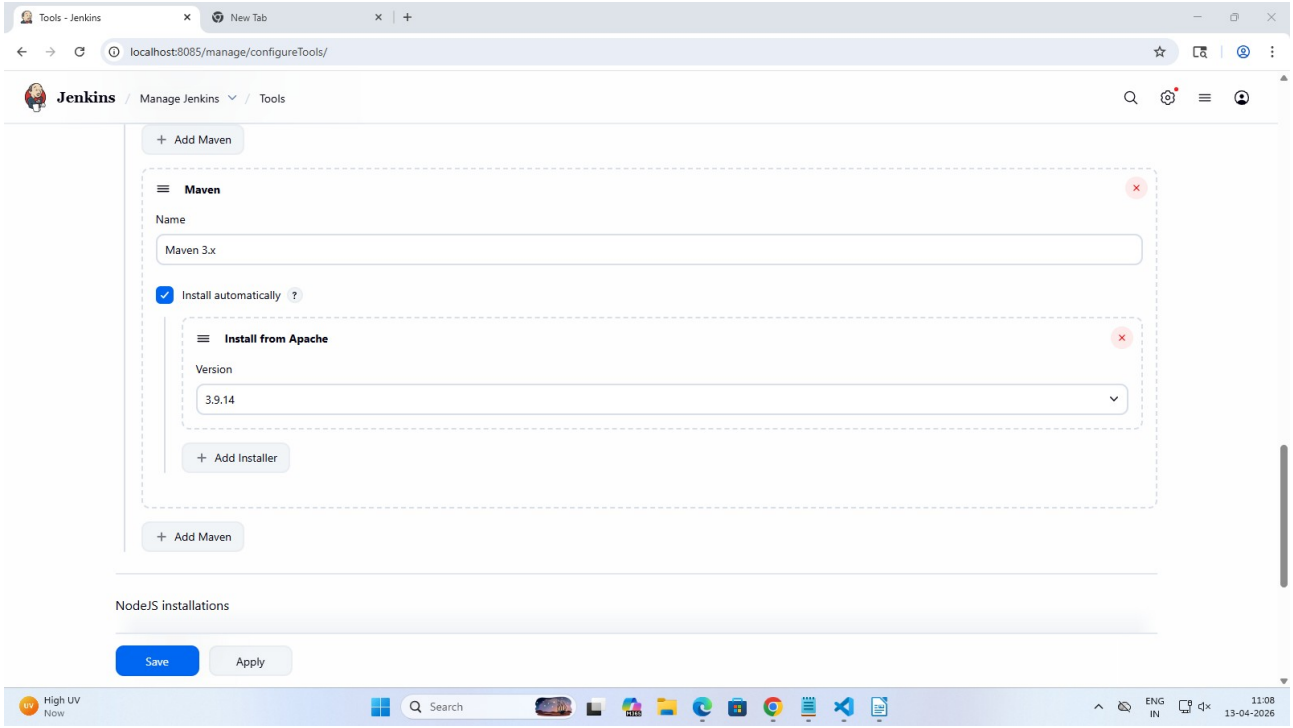
#9 10:55 AM

#8 10:55 AM

Stage View

	Declarative: Checkout SCM	Declarative: Tool Install	Build	Test	Package
Average stage times: (full run time: ~16s)	1s	728ms	3s	3s	4s
#10 Apr 13 11:06 1 commit	1s	226ms	2s	3s	3s
#9 Apr 13 10:55 No Changes	1s	219ms	2s	3s	4s
#8 Apr 13 10:55 No Changes	1s	271ms	3s	3s	3s
#7 Apr 13 10:55 No Changes	1s	300ms	2s	3s	3s
#6 Apr 13 10:54 No Changes	1s	167ms	2s	3s	3s
#5 Apr 13 1 commit	1s	364ms	8s	7s	6s

Hot weather Now | Search | ENG IN | 11:07 13-04-2026



Codes

Voter.java

```
public class Voter {
  private String name;
  private int age;
  private String citizenship;
  private boolean hasValidID;

  // Constructor: Initializes the voter details
  public Voter(String name, int age, String citizenship, boolean hasValidID) {
    this.name = name;
    this.age = age;
    this.citizenship = citizenship;
    this.hasValidID = hasValidID;
  }

  // Business Logic: Determines eligibility and returns the specific reason if
ineligible
  public String checkEligibility() {
    if (age < 18) {
      return "Ineligible: Underage (Must be 18+).";
    }
    if (!citizenship.equalsIgnoreCase("Indian")) {
      return "Ineligible: Not an Indian citizen.";
    }
```

```
    if (!hasValidID) {
      return "Ineligible: Voter ID is invalid or missing.";
    }
    return "Eligible to vote!";
  }

  public String getName() { return name; }
}
```

MainApp.java

```
import java.util.Scanner;

public class MainApp {
  public static void main(String[] args) {
    Scanner scanner = new Scanner(System.in);

    System.out.println("=== Voting Eligibility Checker ===");

    System.out.print("Enter Voter Name: ");
    String name = scanner.nextLine();

    System.out.print("Enter Age: ");
    int age = scanner.nextInt();

    System.out.print("Enter Citizenship (e.g., Indian): ");
    String citizen = scanner.next();

    System.out.print("Does the voter have a valid ID? (true/false): ");
    boolean idStatus = scanner.nextBoolean();


    Voter person = new Voter(name, age, citizen, idStatus);

    // Print the final result
    System.out.println("\n--- Result for " + person.getName() + " ---");
    System.out.println(person.checkEligibility());

    scanner.close();
  }
}
```

VoterTest.java

```
import static org.junit.jupiter.api.Assertions.*;
import org.junit.jupiter.api.Test;
```

```
public class VoterTest {

  @Test
  void testEligibleVoter() {
    Voter v = new Voter("Raj", 25, "Indian", true);
    assertEquals("Eligible to vote!", v.checkEligibility());
  }

  @Test
  void testUnderageVoter() {
    Voter v = new Voter("Sara", 17, "Indian", true);
    assertEquals("Ineligible: Underage (Must be 18+).", v.checkEligibility());
  }

  @Test
  void testForeignCitizen() {
    Voter v = new Voter("John", 30, "Canadian", true);
    assertEquals("Ineligible: Not an Indian citizen.", v.checkEligibility());
  }

  @Test
  void testInvalidID() {
    Voter v = new Voter("Ali", 22, "Indian", false);
    assertEquals("Ineligible: Voter ID is invalid or missing.",
v.checkEligibility());
  }
}
```

pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
     xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
     xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
http://maven.apache.org/xsd/maven-4.0.0.xsd">
  <modelVersion>4.0.0</modelVersion>

  <groupId>com.voting</groupId>
  <artifactId>VotingSystem</artifactId>
  <version>1.0-SNAPSHOT</version>

  <properties>
    <maven.compiler.source>17</maven.compiler.source>
    <maven.compiler.target>17</maven.compiler.target>
  </properties>

  <dependencies>
    <dependency>
      <groupId>org.junit.jupiter</groupId>
      <artifactId>junit-jupiter-api</artifactId>
```

```
      <version>5.10.0</version>
      <scope>test</scope>
    </dependency>
  </dependencies>
</project>
```

PipeLine Code -

```
pipeline {
  agent any

  tools {
    // Ensure this name matches exactly what you set in Jenkins Tools
    maven 'Maven 3.x'
  }

  stages {
    stage('Build') {
      steps {
        // Changed 'sh' to 'bat' for Windows compatibility
        bat 'mvn clean compile'
      }
    }

    stage('Test') {
      steps {
        // Changed 'sh' to 'bat'
        bat 'mvn test'
      }
    }

    stage('Package') {
      steps {
        // Changed 'sh' to 'bat'
        bat 'mvn package'
      }
    }
  }
}
```